

Arterial function of carotid and brachial arteries in postmenopausal vegetarians

Background: Vegetarianism is associated with a lower risk of cardiovascular disease. However, studies of arterial function in vegetarians are limited. **Methods:** This study investigated arterial function in vegetarianism by comparing 49 healthy postmenopausal vegetarians with 41 age-matched omnivores. The arterial function of the common carotid artery was assessed by carotid duplex, while the pulse dynamics method was used to measure brachial artery distensibility (BAD), compliance (BAC), and resistance (BAR). Fasting blood levels of glucose, lipids, lipoprotein (a), high-sensitivity C-reactive protein, homocysteine, and vitamin B12 were also measured. **Results:** Vegetarians had significantly lower serum cholesterol, high-density and low-density lipoprotein, and glucose compared with omnivores. They also had lower vitamin B12 but higher homocysteine levels. Serum levels of lipoprotein (a) and high-sensitivity C-reactive protein were no different between the two groups. There were no significant differences in carotid beta stiffness index, BAC, and BAD between the two groups even after adjustment for associated covariates. However, BAR was significantly lower in vegetarians than in omnivores. Multiple linear regression analysis revealed that age and pulse pressure were two important determinants of carotid beta stiffness index and BAD. Vegetarianism is not associated with better arterial elasticity. **Conclusion:** Apparently healthy postmenopausal vegetarians are not significantly better in terms of carotid beta stiffness index, BAC, and BAD, but have significantly decreased BAR than omnivores. Prevention of vitamin B12 deficiency might be beneficial for cardiovascular health in vegetarians.